

Alpilla 600 MW Combined Cycle Power Plant

HVAC Design Criteria

Document number	ALP-EPC-00000-HV-DBR-0001
Revision / status	Rev A - IFR, issued 2026-12-14
Originator	Istanbul EPC Mühendislik ve Taahhüt A.Ş.
Discipline / type	hvac / DBR Design basis report
Review class	approval
Basis	requirement:ER-10.06

Rev	Purpose	Date	Description	Prepared	Checked	Approved
A	IFR	2026-12-14	First issue for Owner approval (Basic Design Package)	EPCE	EPC HVAC lead	EPC engineering manager

1. Purpose and scope

1.1 This document defines the design criteria of the heating, ventilation and air conditioning (HVAC) systems of the Alpilla 600 MW combined cycle plant: outdoor and indoor design conditions, air quality and filtration for the coastal atmosphere, pressurisation, redundancy, noise, and fire and smoke provisions. It covers the HVAC package 00SAA (chillers, chilled water pumps, air handling units, pressurisation units, hall ventilators and battery room exhaust) and the ventilation of the buildings of the plant.

1.2 Not included: the ventilation and fire protection of the gas turbine and steam turbine enclosures and of the HRSG (supplied by the power island OEM), the process ventilation of the electro-chlorination hydrogen dilution (equipment 00PBA10, process discipline) and hazardous area ventilation of the gas station; the interfaces shall be recorded in the interface register.

2. Governing requirements

ER-10.06: HVAC for all buildings; electrical and control rooms with N+1 cooling and positive pressure with filtered (coastal) air.

ER-02.02: Equipment shall operate at full capability between -8 degC and +40 degC dry bulb and 18 % to 100 % relative humidity, and survive -15 degC to +45 degC. Climatic data per ALP-OWN-MET-001 (10-year site record; extremes -11.4 degC / +39.6 degC).

ER-02.08: Outdoor equipment and structures shall be designed for corrosivity category C5 (ISO 12944-2) with a coating system of high durability (H, more than 15 years to first major maintenance).

ER-10.05: Fire detection and protection per NFPA 850 and Turkish Fire Regulation: ring main, 2x100 % electric plus diesel fire pumps, water mist/CO2 for GT enclosure, deluge for transformers, gas detection, fire alarm system.

ER-14.03: Maximum 85 dB(A) at 1 m from any equipment in normal operation.

3. Codes and standards

3.1 HVAC design shall follow ASHRAE Handbook and Standards (62.1 for ventilation, 55 for comfort, 90.1 for energy) or EN 16798-1 where more onerous, EN 12831 for heating loads, EN 14511 for chiller rating, EN ISO 16890 for particulate filters, EN 1822 for EPA filters, EN 50272-2 for battery rooms, NFPA 90A for air distribution fire and smoke control, NFPA 850 and NFPA 72 for fire protection and detection and the Turkish Fire Regulation for buildings.

4. Outdoor design conditions

Parameter	Value	Condition	Status	Basis
Ambient temperature - maximum for full capability	40 degC		confirmed	ER-02.02
Ambient temperature - minimum for full capability	-8 degC		confirmed	ER-02.02
Ambient temperature - survival range	-15 to +45 degC		confirmed	ER-02.02
Annual mean temperature (measured)	14.9 degC	2015-2024 site mast	confirmed	SRC-OWNR-0001

Parameter	Value	Condition	Status	Basis
Chloride deposition maximum	420 mg/m ² /day	ISO 9225, category S3	confirmed	SRC-OWNR-0001, ER-02.10
Ground flash density	2.4 flashes/km ² /year	26 thunderstorm days/year	confirmed	SRC-OWNR-0001, ER-02.12
Performance test ambient range	5 to 30 degC		confirmed	SRC-OWNR-0001, ER-02.11
PM10 maximum (event)	260 ug/m ³	Saharan dust episodes	confirmed	SRC-OWNR-0001, ER-02.10
Record maximum / minimum temperature	39.6 / -11.4 degC	1960-2024	confirmed	SRC-OWNR-0001
Sea-salt aerosol annual mean	14 ug/m ³	as NaCl	confirmed	SRC-OWNR-0001
Sea-salt aerosol maximum (event)	165 ug/m ³	as NaCl, lodos storms	confirmed	SRC-OWNR-0001, ER-02.10
SRC ambient dry-bulb temperature	15 degC	Site Reference Conditions	confirmed	ER-02.03
SRC barometric pressure	1,011.5 mbar	Site Reference Conditions	confirmed	ER-02.03
SRC relative humidity	70 %	Site Reference Conditions	confirmed	ER-02.03
Summer design dry bulb (0.4 %)	32.8 degC	coincident RH 48 %	confirmed	SRC-OWNR-0001, ER-02.11
Winter design dry bulb (99.6 %)	-4.8 degC	coincident RH 88 %	confirmed	SRC-OWNR-0001, ER-02.11

4.1 The cooling design shall be based on the maximum ambient dry-bulb temperature for full capability of 40 degC (DP-CASE-0008) and the site summer design condition (0.4 %, DP-OWNR-0001) for comfort cooling load calculations of buildings with a 10 % safety margin; equipment (air-cooled chillers, condensers, fans) shall be rated to give the required capacity at 40 degC and shall operate without damage at the survival range (-15 degC to +45 degC, DP-CASE-0009); the capacity derating between 40 degC and 45 degC shall be stated by the chiller supplier.

4.2 The heating design shall be based on the winter design temperature of DP-OWNR-0002 (-4.8 degC at 88 % RH) for comfort heating and on -8 degC (DP-CASE-0007) for freeze protection and the minimum room temperatures of the process buildings; the design shall be checked at the record minimum of -11.4 degC (DP-OWNR-0003) without freezing of water piping.

4.3 The atmosphere is marine C5 with sea-salt aerosol, chloride deposition and Saharan dust (DP-OWNR-0005 to DP-OWNR-0008); all outdoor equipment (air-cooled chillers, condenser coils, fans, dampers, ducts, louvres, roof ventilators) shall be suited to it (section 13).

5. Indoor design conditions

Room type	Design condition	Remarks
Central control room and control building offices	22 degC +/- 2 K, RH 40 to 60 %, non-condensing, all year	Occupied; fresh air per section 6 [ER-10.06]
Electronic / server / DCS cabinet rooms	22 degC +/- 2 K, RH 40 to 55 %; maximum rate of change 5 K/h	Unmanned; 24 h operation [ER-10.06]
Electrical rooms - MV and LV switchgear, transformers, UPS	maximum 35 degC, minimum 10 degC, RH <= 70 % non-condensing	Equipment rated 40 degC; a 5 K margin is kept [ER-10.06]
Battery rooms	20 to 25 degC (cell temperature), maximum 30 degC; hydrogen concentration < 1 vol % (25 % of the LEL)	Section 8 [ER-10.06]
GIS building	5 to 35 degC, RH <= 75 %	SF6 low-level extraction [ER-10.06]
Gas and steam turbine halls	maximum ambient + 10 K in summer, minimum +5 degC in winter	Ventilation by roof ventilators; heat gain from OEM [ER-10.06, DP-CASE-0008]
Workshop and warehouse	summer maximum 32 degC (ventilation), winter minimum 12 degC; office part 24 degC +/- 2 K	[ER-10.06]
Administration building offices, meeting rooms	summer 24 degC +/- 2 K, winter 21 degC +/- 2 K, RH 30 to 60 %	[ER-10.06]
Water treatment, pump and chemical buildings	ventilation only: maximum ambient + 8 K; minimum +5 degC; chemical rooms with dedicated exhaust	[ER-10.06]
Gatehouse and fire station	summer 26 degC, winter 20 degC	[ER-10.06]

6. Design methodology and air quality

6.1 Cooling and heating loads shall be calculated with a transient method (or the ASHRAE heat balance method) and include: equipment heat release from the electrical load list (heat losses of transformers, switchgear, UPS, chargers, drives, DCS cabinets), lighting, occupants, solar and conduction gains and fresh air; a 10 % design margin and a 5 % allowance for future additions shall be added to the calculated loads; the equipment heat release shall be taken from the equipment records and the electrical load list, not retyped.

6.2 Fresh air shall be provided at not less than 10 l/s per person for occupied rooms (7 l/s per person plus 0.7 l/s per m² as ASHRAE 62.1 minimum) and enough to maintain the pressurisation of section 7; toilets and kitchens shall be exhausted at not less than 10 air changes per hour or 25 l/s per cubicle.

6.3 Air distribution: main duct velocity not more than 8 m/s, branch ducts 5 m/s, occupied-zone outlets 3 m/s; friction rate not more than 1 Pa/m for low pressure ducts; ductwork shall be leak class C (EN 12237 / DW/144 class B or better) and leakage shall be tested.

7. Filtration and pressurisation of buildings

7.1 Filtration against salt and dust: the outdoor air intakes of all conditioned buildings shall have weather louvres with drift eliminators (removal of at least 99 % of droplets above 10 microns at the design face velocity of 2.5 m/s or less), followed by a G4 pre-filter and an F7 filter; the control building, electronic rooms, MV / LV rooms, UPS and battery rooms and the GIS building shall have a final F9 filter (EN ISO 16890 ePM1 $\geq$ 80 %), and the control building and electronic rooms shall in addition have an E10 EPA filter (EN 1822) or gas-phase filtration to keep the airborne chloride and sulphur gases at the ISA 71.04 severity G1 level.

7.2 The filters shall be sized for the maximum sea-salt aerosol event (165 ug/m³, DP-OWNR-0005) and the PM10 episode (260 ug/m³, DP-OWNR-0008) with a filter change interval of at least 6 months at the annual mean salt concentration (14 ug/m³, DP-OWNR-0006); each filter bank shall have differential pressure monitoring with alarm at 2 times the clean value and a bypass-free arrangement (sealed frames); the filters shall be accessible and replaceable without interrupting the pressurisation of the room (see redundancy).

7.3 Pressurisation: control and electronic rooms and MV / LV / UPS / battery-adjacent electrical rooms shall be kept at a positive pressure of not less than 25 Pa (50 Pa for the control building and the electronic rooms) relative to the outside and to adjacent unconditioned spaces; the airflow at open doors shall be enough to keep 0.5 m/s in the door opening at the standard leakage of the building; the pressure shall be controlled by a pressure sensor per room (dual sensors on the control building) with modulating supply and relief dampers; loss of pressure shall raise an alarm in the control room.

7.4 The outdoor air intakes shall be located at not less than 3 m above grade in non-hazardous areas (outside the classified zones of HS-HAZ-0008) on the upwind side of the fuel gas station, the LDO and ammonia storage and the stack, and shall be equipped with combustible gas detectors (2oo3 voting) that close the intake dampers and switch the unit to recirculation at 10 % LEL and with an ammonia detector for the areas near the SCR storage, so that hazardous gases cannot be drawn into the pressurised buildings.

8. Battery rooms and hydrogen dilution

8.1 Battery rooms shall be ventilated by two exhaust fans of 100 % capacity (duty / standby, 3,000 m³/h each per the equipment records) discharging directly to the outside at a safe location at least 1.5 m above the roof and away from air intakes; the air flow shall keep the hydrogen concentration below 1 vol % (25 % of the LEL of 4 vol %) at the maximum charging current calculated per EN 50272-2, and the required airflow shall be verified for the final battery type and capacity from the battery datasheet.

8.2 The battery room ventilation shall be monitored by an airflow switch; on loss of ventilation the battery charger shall change to float voltage (boost charging inhibited) and an alarm shall be raised; hydrogen detectors shall alarm at 10 % LEL and trip the charger at 25 % LEL; the fans, motors and detectors shall be certified for Zone 2 IIC T1 in the room and the room shall be considered per IEC 60079-10-1; no ignition source or non-certified equipment is permitted in the room.

9. Turbine halls and process buildings

9.1 Turbine halls (gas turbine hall and steam turbine hall) shall be ventilated by roof ventilators (4 x 25 % groups per hall per the equipment records) with low-level louvred inlet openings fitted with salt drift eliminators and coarse filters; the airflow shall limit the hall temperature to ambient + 10 K at 40 degC with the heat release from the OEM (radiation and leakage of casings, generators, lube oil systems) and shall give at least 6 air changes per hour at the base ventilation, with the failure of one 25 % group not causing the design temperature to be exceeded by more than 3 K.

9.2 The gas turbine hall and the fuel gas areas shall not depend on the HVAC for the safety ventilation of the GT enclosure (OEM); where the hall ventilation is used for the dilution of gas releases the ventilation shall be in operation continuously, the gas detection of the fire and gas system shall trip fuel gas supply and start the fans in high speed, and the hazardous area classification shall take credit only for the ventilation availability of the design.

9.3 The GIS building shall be ventilated with low-level extraction (SF₆ is heavier than air) with oxygen depletion / SF₆ monitors at floor level, at least 4 air changes per hour under normal operation and 8 air changes per hour in the emergency mode started by the SF₆ monitor; entry warning signs and ventilation before entry shall be provided.

9.4 Workshop, warehouse, water treatment and chemical rooms shall be ventilated by exhaust with make-up air from filtered louvred inlets; the chemical rooms (hypochlorite, acids, caustic, ammonia dosing) shall have dedicated exhaust of at least 10 air changes per hour with the discharge above roof level, corrosion-resistant (GRP / PP) ducts and fans, and shall be negative in pressure relative to adjacent rooms.

10. Chilled water plant and redundancy

10.1 The chilled water plant shall be N+1: three air-cooled screw chillers of 50 % duty each (3 x 50 %; two chillers in duty and one standby, 480 kW cooling each at 7/13 degC per the equipment records), two chilled water pumps 2 x 100 % and the associated valves and controls arranged so that the loss of any one chiller, pump or power feeder does not reduce the cooling of the electrical and control rooms; the chillers shall be supplied from different MCC sections (both sections of the essential distribution).

10.2 Air handling units serving the control room, the electronic rooms, the MV and LV rooms and the UPS rooms shall have N+1 provisions in fans (a fan wall with n+1 fans or duty / standby fans with automatic changeover), power supply and controls, so that a single failure does not interrupt the cooling and the pressurisation; the coils may be single provided they are split in two independent circuits; the pressurisation fan of the control building shall have a standby fan (equipment records 00SAA11AN001 to 005 and 008: 2 x 100 % supply fans).

10.3 Power supply of the HVAC of the control and electrical rooms shall be from the essential (emergency diesel backed) section of the auxiliary system so that the cooling and pressurisation is restored automatically after the transfer to the emergency diesel generator within the time of the ride-through of the rooms (not more than 5 minutes with a room temperature rise of not more than 3 K); the chilled water thermal storage or the inertia of the loop shall be shown to cover the restart time of the chillers.

10.4 The chillers and pumps shall be selected for the maximum ambient of 40 degC, with the refrigerant a low-GWP fluid (R1234ze or R513A) and the coil fins epoxy or phenolic coated for C5; the air intakes and the layout shall avoid hot air recirculation and shall have a clearance for the wind (design wind of DP-CASE-0004) and shall be protected against salt spray by a wall or the layout.

Tag	Description	Service	Redundancy	Rating	Design P / T	Material
00SAA10	HVAC plant (chillers, AHUs, pressurisation)	HVAC plant package: chillers, chilled water pumps, AHUs, pressurisation (skid/package parent)	Chillers 3 x 50 %	960 kW cooling (2 chillers duty), 450 kW, 400 V		Carbon steel piping, insulated
00SAA10AN001	Chiller A	Chilled water chiller A (air-cooled screw, 7/13 degC)	3 x 50 %	480 kW cooling, 150 kW, 400 V	10 barg / 50 degC	Evaporator CS/SS; condenser Cu/Al coils coated for C5 marine
00SAA10AN002	Chiller B	Chilled water chiller B (air-cooled screw, 7/13 degC)	3 x 50 %	480 kW cooling, 150 kW, 400 V	10 barg / 50 degC	Evaporator CS/SS; condenser Cu/Al coils coated for C5 marine
00SAA10AN003	Chiller C	Chilled water chiller C (air-cooled screw, 7/13 degC)	3 x 50 %	480 kW cooling, 150 kW, 400 V	10 barg / 50 degC	Evaporator CS/SS; condenser Cu/Al coils coated for C5 marine
00SAA10AP001	Chilled water pump A	Chilled water circulation pump	2 x 100 %	150 m3/h, 35 m, 22 kW, 400 V	8 barg / 50 degC	Cast iron / bronze impeller, insulated CS piping
00SAA10AP002	Chilled water pump B	Chilled water circulation pump	2 x 100 %	150 m3/h, 35 m, 22 kW, 400 V	8 barg / 50 degC	Cast iron / bronze impeller, insulated CS piping
00SAA11AN001	AHU - control room	AHU - control room	1 x 100 % unit with 2 x 100 % supply fans (N+1)	30,000 m3/h, 15 kW, 400 V	0.1 barg / 50 degC	Galvanised/epoxy-coated steel casing, C5
00SAA11AN002	AHU - electronic rooms	AHU - electronic rooms	1 x 100 % unit with 2 x 100 % supply fans (N+1)	30,000 m3/h, 15 kW, 400 V	0.1 barg / 50 degC	Galvanised/epoxy-coated steel casing, C5
00SAA11AN003	AHU - electrical building MV rooms	AHU - electrical building MV rooms	1 x 100 % unit with 2 x 100 % supply fans (N+1)	22,000 m3/h, 11 kW, 400 V	0.1 barg / 50 degC	Galvanised/epoxy-coated steel casing, C5
00SAA11AN004	AHU - electrical building LV rooms	AHU - electrical building LV rooms	1 x 100 % unit with 2 x 100 % supply fans (N+1)	22,000 m3/h, 11 kW, 400 V	0.1 barg / 50 degC	Galvanised/epoxy-coated steel casing, C5
00SAA11AN005	AHU - GIS building	AHU - GIS building	1 x 100 % unit with 2 x 100 % supply fans (N+1)	15,000 m3/h, 7.5 kW, 400 V	0.1 barg / 50 degC	Galvanised/epoxy-coated steel casing, C5
00SAA11AN006	AHU - administration building	AHU - administration building	1 x 100 %	22,000 m3/h, 11 kW, 400 V	0.1 barg / 50 degC	Galvanised/epoxy-coated steel casing, C5
00SAA11AN007	AHU - workshop	AHU - workshop	1 x 100 %	15,000 m3/h, 7.5 kW, 400 V	0.1 barg / 50 degC	Galvanised/epoxy-coated steel casing, C5
00SAA11AN008	Pressurisation unit - control building	Pressurisation unit - control building	1 x 100 % unit with 2 x 100 % supply fans (N+1)	16,000 m3/h, 5.5 kW, 400 V	0.1 barg / 50 degC	Galvanised/epoxy-coated steel casing, C5

Tag	Description	Service	Redundancy	Rating	Design P / T	Material
00SAA12AN001	GT hall roof ventilator A	GT hall roof ventilator A	4 x 25 %	110,000 m ³ /h, 15 kW, 400 V	0 barg / 60 degC	Aluminium/GRP impeller, coated housing, C5
00SAA12AN002	GT hall roof ventilator B	GT hall roof ventilator B	4 x 25 %	110,000 m ³ /h, 15 kW, 400 V	0 barg / 60 degC	Aluminium/GRP impeller, coated housing, C5
00SAA12AN003	GT hall roof ventilator C	GT hall roof ventilator C	4 x 25 %	110,000 m ³ /h, 15 kW, 400 V	0 barg / 60 degC	Aluminium/GRP impeller, coated housing, C5
00SAA12AN004	GT hall roof ventilator D	GT hall roof ventilator D	4 x 25 %	110,000 m ³ /h, 15 kW, 400 V	0 barg / 60 degC	Aluminium/GRP impeller, coated housing, C5
00SAA13AN001	ST hall roof ventilator A	ST hall roof ventilator A	4 x 25 %	110,000 m ³ /h, 15 kW, 400 V	0 barg / 60 degC	Aluminium/GRP impeller, coated housing, C5
00SAA13AN002	ST hall roof ventilator B	ST hall roof ventilator B	4 x 25 %	110,000 m ³ /h, 15 kW, 400 V	0 barg / 60 degC	Aluminium/GRP impeller, coated housing, C5
00SAA13AN003	ST hall roof ventilator C	ST hall roof ventilator C	4 x 25 %	110,000 m ³ /h, 15 kW, 400 V	0 barg / 60 degC	Aluminium/GRP impeller, coated housing, C5
00SAA13AN004	ST hall roof ventilator D	ST hall roof ventilator D	4 x 25 %	110,000 m ³ /h, 15 kW, 400 V	0 barg / 60 degC	Aluminium/GRP impeller, coated housing, C5
00SAA14AN001	Battery room exhaust fan A	Battery room exhaust fan A (hydrogen dilution)	2 x 100 %	3,000 m ³ /h, 1.5 kW, 400 V	0 barg / 50 degC	Non-sparking FRP/ATEX-suitable
00SAA14AN002	Battery room exhaust fan B	Battery room exhaust fan B (hydrogen dilution)	2 x 100 %	3,000 m ³ /h, 1.5 kW, 400 V	0 barg / 50 degC	Non-sparking FRP/ATEX-suitable

11. Noise

11.1 The near-field noise of any HVAC equipment shall not exceed 85 dB(A) at 1 m (ER-14.03) and the contribution of the HVAC to the plant noise at the nearest receptor shall be included in the plant noise model (HS-STU-0001) so that the receptor limits at night (DP-CASE-0045) and during the day (DP-EPCE-0010) are met; the roof ventilators and chillers shall have attenuators and acoustic screens where required by the noise study, and a maximum sound power level per unit shall be set in the requisition from the noise study.

Parameter	Value	Condition	Status	Basis
Near-field noise	85 dB(A)	1 m from equipment	confirmed	ER-14.03
Noise at nearest receptor (day)	55 dB(A)	LAeq day	confirmed	SRC-OWNR-0007, ER-14.04
Noise at nearest receptor (night)	45 dB(A)	LAeq night	confirmed	ER-14.04

Item	Value	Unit	Status	Basis / remarks
Room noise - control room and offices	<= 40 dB(A) (control room), <= 45 dB(A) (offices, control room ancillary)		assumption	ER-14.03
Room noise - electrical rooms and battery rooms	<= 65 dB(A)		assumption	ER-14.03
Room noise - workshops, halls	<= 80 dB(A) inside, hearing protection zones marked above 85 dB(A)		assumption	ER-14.03

12. Fire and smoke

12.1 Fire dampers shall be installed where ducts pass through fire compartment boundaries of the Fire Regulation and NFPA 850 with a rating not less than that of the partition (minimum 90 minutes for electrical rooms and 120 minutes for the boundary of the control building), be tested to the relevant standard, be accessible for inspection and fitted with position feedback; ductwork in fire rated shafts may replace the dampers when tested as rated.

12.2 Smoke detection in the return / supply air of every AHU above 2,000 m³/h and in the room shall stop the fans and close the dampers automatically (NFPA 90A, NFPA 72), except the pressurisation units of the escape stairs and the smoke control fans, which continue in the smoke mode; the fire alarm system shall have a manual override for the fire brigade at the fire station and at the control room.

12.3 Rooms protected by gaseous extinguishing (control room floor voids, electronic rooms, cable rooms, GT enclosure per the fire protection concept, CO₂ / inert gas per NFPA 12) shall have their ventilation dampers closing before the discharge and a post-discharge purge extraction of at least 6 air changes per hour started manually after the fire brigade has confirmed the extinguishing; the room shall not be pressurised during the discharge.

12.4 Smoke extraction shall be provided for the corridors and escape stairs of the control, electrical and administration buildings (mechanical smoke extraction at 10 air changes per hour or natural smoke vents of 1 m² aerodynamic area per 100 m² of floor) with make-up air from outside, and for the cable basement and cable rooms by manual smoke extraction

after the extinguishing; the smoke control fans shall be rated for 400 degC for 2 hours, supplied from the essential section and not subject to trips by the HVAC control.

13. Materials, controls and maintainability

13.1 Materials: supply and return ductwork indoors of galvanised sheet steel; outdoor ductwork, louvres and casings of stainless steel 316L or aluminium with a marine coating system, or hot-dip galvanised with paint to ISO 12944 C5 H per ALP-EPC-00000-ME-SPC-0002; fan and motor casings with a marine coating and IP55 minimum protection (IP56 for outdoor motors); chilled water pipes of carbon steel with a paint system per the painting specification and cellular glass or closed cell elastomeric insulation with an aluminium jacket outdoors.

13.2 Controls: the HVAC shall have a local controller per package with a serial link to the DCS (alarms for chiller trip, low pressure of a room, filter differential pressure, fan failure, high room temperature 3 K above the set point and smoke detection), automatic changeover of standby equipment, remote start and stop from the control room for the essential ventilation, and time delay restart after a power failure; the room temperature and pressure signals of the control and electrical rooms shall be hard-wired to the DCS.

13.3 Maintainability: filters, coils, fans and dampers shall be accessible from the platform or the floor with 1.0 m clear service space; equipment on roofs shall have access with guard rails and permanent lifting facilities for units heavier than 25 kg; a spare filter set for one year shall be included in the spare parts list.

References

Codes and standards cited in this document (Codes, Standards and Regulations Register ALP-EPC-00000-GE-LST-0001; latest edition at contract date unless stated).

Code	Title
Fire Regulation	Regulation on Fire Protection of Buildings (Binalarin Yangindan Korunmasi Hakkinda Yonetmelik)
EN 1822	High efficiency air filters (EPA, HEPA and ULPA)
NFPA 12	Standard on Carbon Dioxide Extinguishing Systems
NFPA 850	Recommended Practice for Fire Protection for Electric Generating Plants and HVDC Converter Stations
NFPA 72	National Fire Alarm and Signaling Code
IEC 60079-10-1	Explosive atmospheres - Classification of areas - Explosive gas atmospheres
ISO 12944-2	Paints and varnishes - Corrosion protection of steel structures by protective paint systems - Part 2: Classification of environments